

```
$ mysql -e "SHOW CREATE TABLE orders" --normalize
```



// 1NF → 2NF → 3NF □□□□□

1NF

□□□

2NF

□□□□□□

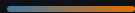
3NF

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OLAP □□□

\$ cat schema-plan.md



01 / □□

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02 / □□□□

1NF □□□ → 2NF → 3NF

03 / □□

JOIN □□□

04 / □□□□

OLTP vs OLAP □□□

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PART 01



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order_id	customer_name	customer_email	product_name	price
1	□□	<u>zhang@example.com</u>	□□□□□	30000
2	□□	<u>zhang@example.com</u>	□□	500
3	□□	<u>li@example.com</u>	□□	1200

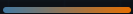
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□□□□□□ email □□□□

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□□□ email□□ UPDATE □□□□

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ROOT CAUSE

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CHECKPOINT_01

データベースの UPDATE 37 回実行後 3 分経過

A データが更新された

B データが削除された

C データが追加された

PART 02



1NF → 2NF → 3NF →

1NF□□□□□□□□

BEFORE — □□□□□

student	courses
□□	□□□, □□□, □□

AFTER — 1NF

student	course
□□	□□□
□□	□□□
□□	□□

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2NF

(order_id, product_id) product_name product_id —

BEFORE —

order_id	product_id	product_name	qty
1	P1		1
2	P1		3

AFTER —

order_id	product_id	qty
1	P1	1
product_id		product_name
P1		

3NF□□□□□□□

student_id → dept_id → dept_name□dept_name □□ dept_id □□□□□□ — □□□□□□□

BEFORE — □□□□

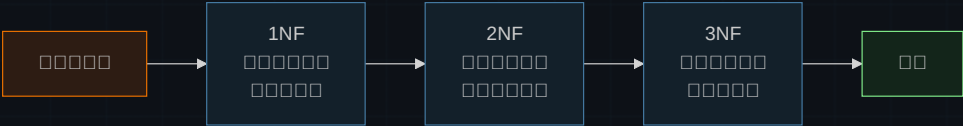
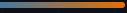
student_id	name	dept_id	dept_name
S1	□□	D1	□□□
S2	□□	D1	□□□

AFTER — □□□□□

student_id	name	dept_id
S1	□□	D1
dept_id	dept_name	
D1	□□□	

□□□□□□□□□□□□ — □□□□□□□□□□□□

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PART 03



□□□□□□JOIN □□

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JOIN □□
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□□ JOIN □□□□□

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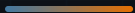
OLTP vs OLAP

OLTP	OLAP	Comparison
Transactional	Analytical	Operational
Real-time	Batch	Reporting
High frequency	Low frequency	Ad-hoc
Simple queries	Complex queries	Aggregations
Point-in-time	Historical	Rollup
OLTP	OLAP	OLAP

STRATEGY

OLTP — OLAP

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1NF □□□ → 2NF □□□□ → 3NF □□□□

3NF □□

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OLTP □□□□OLAP □□□□□□□

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Schema normalized.

XXXXXXXXXXXX JOIN XXXXX

```
$ mysql -e "EXPLAIN SELECT ..."
```

1NF → 2NF → 3NF → XXXXXXXX